FOUNDATION COURSE

START YOUR
PREPARATION
FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 02

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Recap of previous lecture

(POLYNOMIAL & QUADRATIC E&U.)



Topics to be covered

Inequalities & Mod function



Defⁿ- If two algebraic expressions are connected by the symbol " $>, <, \geq, \leq$ " then this expression is called Inequality

for eg $2x+3 = -5x+7$ it is an equation

while $2x+3 > -5x+7$ it is an Inequality

Types- Inequality forms by $>$ or $<$ symbol known as Strict Inequ.

while " " " $\geq$ or $\leq$ " " " SLACK "

Properties-

Properties → (1) if $ab > 0$ or $\frac{a}{b} > 0$ then a & b have same sign

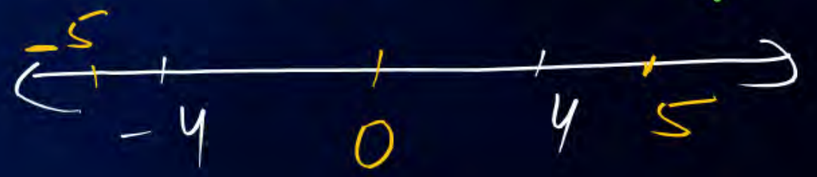
(2) if $ab < 0$ or $\frac{a}{b} < 0$ then a & b are of opposite sign

(3) if we add or subtract same number in an inequality, It will not change

(4) if we Multiply or divide by +ve number " " , " " " "

(5) if " " " " by −ve " " " , then it's sign changes

for eg $\boxed{x > 4} \Rightarrow -x < -4$

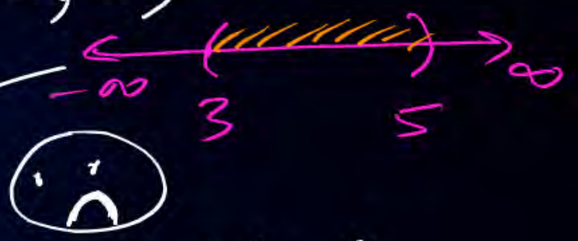


Let $x = 5$ then $-x = -5$

$\because 5 > 4$ but $-5 < -4$

⑥ Let $a<b$ such that $\boxed{(x-a)(x-b)<0 \Rightarrow a<x<b}$ i.e $x \in (a,b)$

for eg $(x-3)(x-5)<0 \Rightarrow 3<x<5$ or $x \in (3,5)$



Proof: Let $x=4 \Rightarrow (4-3)(4-5)=1\times(-1)=-1<0$ ✓

$x=2 \Rightarrow (2-3)(2-5)=(-1)(-3)=+3>0$

$x=6 \Rightarrow (6-3)(6-5)=(3)(1)=3>0$ Not justified

⑦ Let $a<b$ s.t $(x-a)(x-b) \leq 0 \Rightarrow a \leq x \leq b$ i.e $x \in [a,b]$

⑧ Let $a<b$ s.t $\boxed{(x-a)(x-b)>0 \Rightarrow x<a \text{ or } x>b}$ i.e $(-\infty,a) \cup (b,\infty)$

⑨ Let $a<b$ s.t $(x-a)(x-b) \geq 0 \Rightarrow x \leq a$ or $x \geq b$ i.e $(-\infty,a] \cup [b,\infty)$

justification: i.e $(x-2)(x+5)>0 \Rightarrow (x-(-5))(x-2)>0 \Rightarrow x<-5$ or $x>2$



Let us take $x=0$ then $(0-2)(0+5)=-10<0$ i.e Not Satisfied

Linear Inequality in one Variable → it's Sol^{n} can be plotted on Number line

Q. Solve: $\frac{5-2x}{3} \leq \frac{x}{6} - 5$

Sol^{n} $\frac{5-2x}{3} \leq \frac{x-30}{6}$

$\frac{5-2x}{1} \leq \frac{x-30}{2}$

$10 - 4x \leq x - 30$

$10 - 4x - x + 30 \leq 0$

$-5x + 40 \leq 0$

$5x - 40 \geq 0 \Rightarrow x \geq 8$

i.e. $[8, \infty)$ i.e.

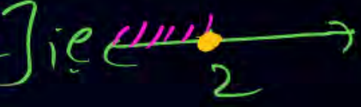


Q. $37 - (3x+5) \geq 9x - 8(x-3)$

$32 - 3x \geq x + 24$

$32 - 3x - x - 24 \geq 0 \Rightarrow -4x + 8 \geq 0$

$4x - 8 \leq 0 \Rightarrow x \leq 2$ i.e. $(-\infty, 2]$ i.e.



Q. $\frac{x}{4} < \left(\frac{5x-2}{3}\right) - \left(\frac{7x-3}{5}\right)$

$\frac{x}{4} < \frac{25x - 10 - 21x + 9}{15}$

$\frac{x}{4} < \frac{4x - 1}{15}$

$15x < 16x - 4$

$-x < -4$

$x > 4$

i.e. $(4, \infty)$



(4) $\frac{x-2}{x+5} > 2$

$\frac{x-2}{x+5} - 2 > 0$

$\frac{x-2-2x-10}{x+5} > 0$

$\frac{-x-12}{x+5} > 0$

$\frac{x+12}{x+5} < 0$

ie $(x+12)(x+5) < 0$

$(x-(-12))(x-(-5)) < 0$

$-12 < x < -5$

(5) $\frac{x}{2x+1} \geq \frac{1}{4}$, $\frac{6x}{4x+1} < \frac{1}{2}$ (No soln)

HW

(6) Solve $3x-7 < 5+x$ & $11-5x \leq 1$

$3x-x < 5+7$ & $-5x \leq 1-11$

$2x < 12$ & $-5x \leq -10$

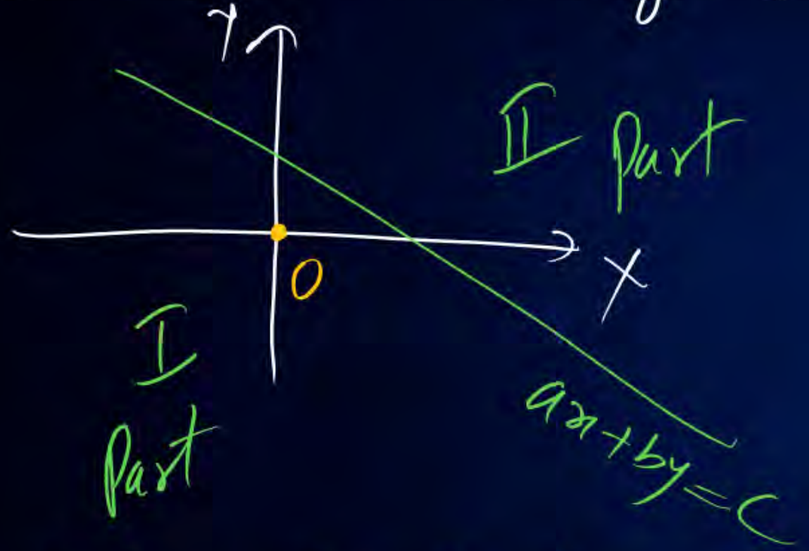
$x < 6$ & $x \geq 2$



ie $2 \leq x < 6$ ie $x \in [2, 6)$

Linear Inequalities in 2 Variables → it is (ax+by) can be plotted on 2D plane

(*) Consider the straight line $ax+by=c$

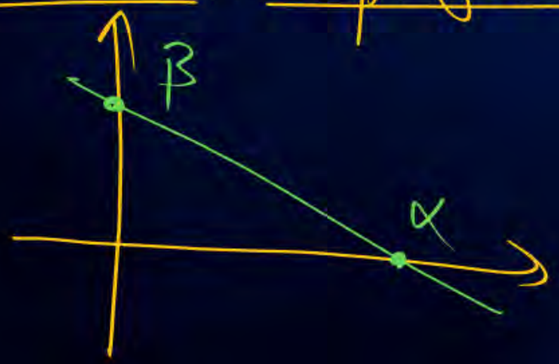


i.e any straight line divides 2D plane into two equal parts. Now two cases arises

Case I: $ax+by>c$ Case II: $ax+by<c$

Note:- Straight line in Intercept form:-

$$\frac{x}{\alpha}+\frac{y}{\beta}=1$$

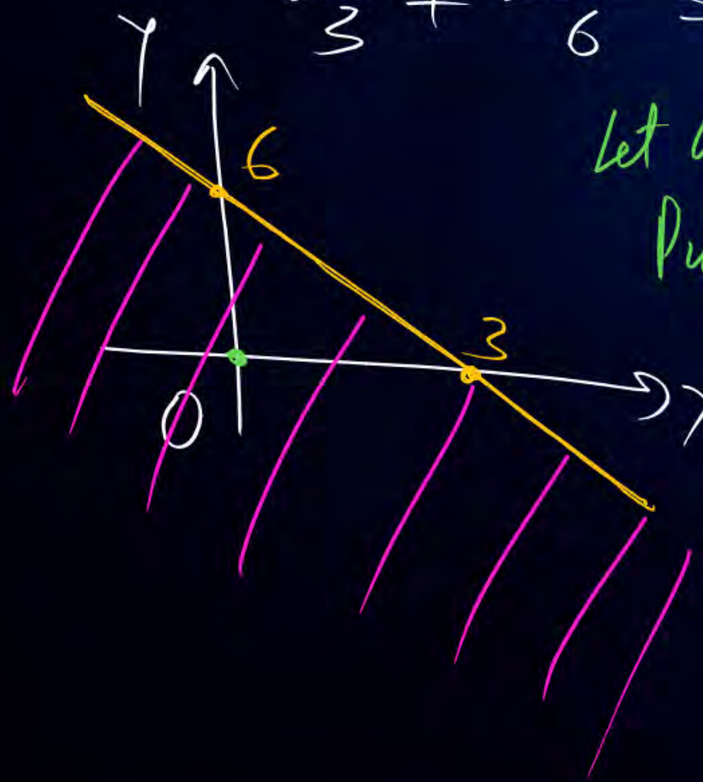


Q. Find the feasible Region for

$40x + 20y \leq 120$ ——— (1)

Sol: $\frac{40x}{120} + \frac{20y}{120} \leq 1$

$\frac{x}{3} + \frac{y}{6} \leq 1$



Let us take (0,0) & Putting this point in (1)

$\therefore 0+0 \leq 120$

i.e $0 \leq 120$

Q. Solve: $3x+2y > 6$ ——— (1)

Sol: $\frac{3x}{6} + \frac{2y}{6} > 1 \Rightarrow \frac{x}{2} + \frac{y}{3} > 1$



Putting (0,0) in (1) we have

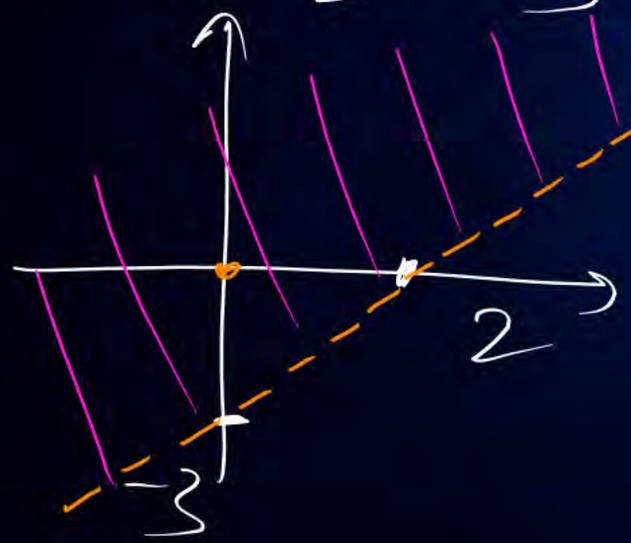
$0+0 > 6$ i.e $0 > 6$

i.e Not Valid

Q. Solve; $-3x+2y>-6$

Sol: $3x-2y<6$ — (1)

or $\frac{x}{2}+\frac{y}{-3}<1$



Put (0,0) in (1)

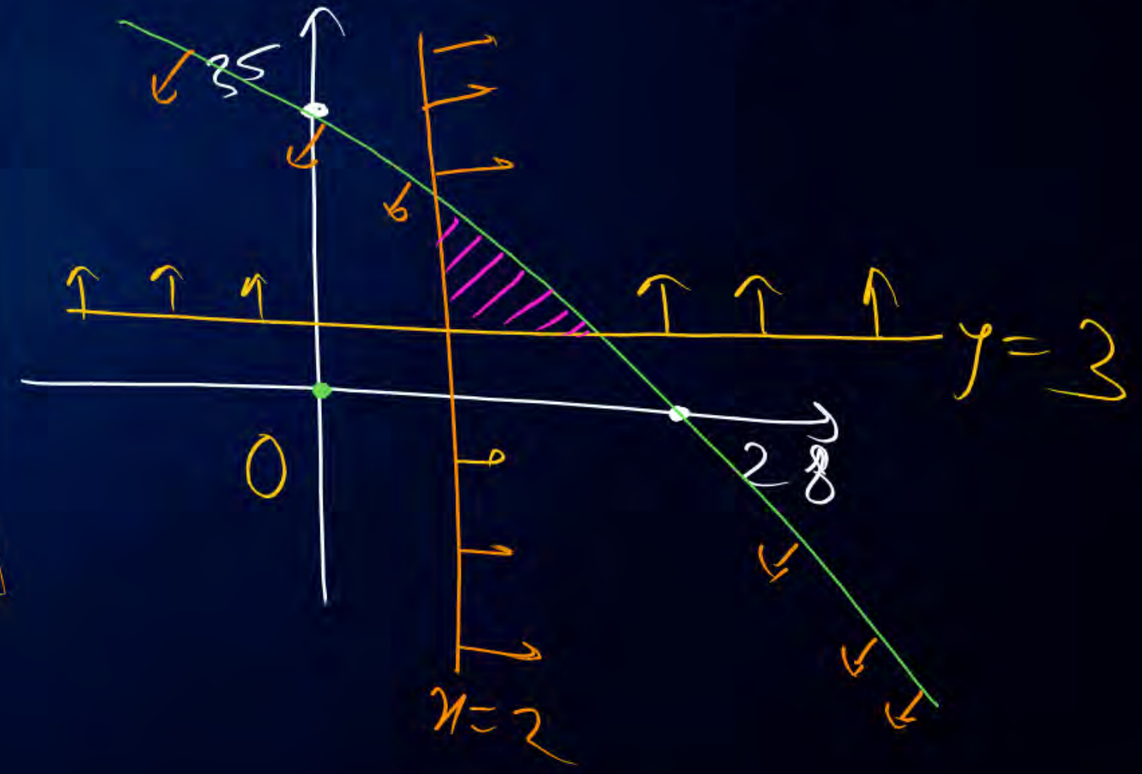
$0-0<6$

i.e. $0<6$ (Valid)

Q. Solve; $5x+4y\leq 140$

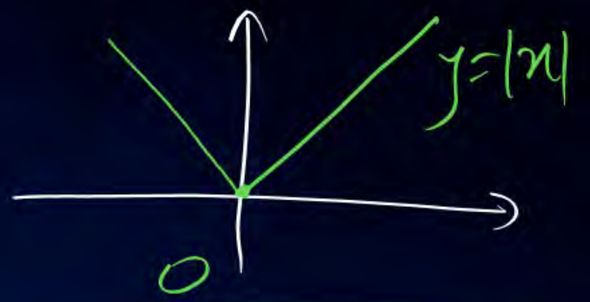
$x\geq 2$, $y\geq 3$

Sol: $\frac{5x}{140}+\frac{4y}{140}\leq 1 \Rightarrow \frac{x}{28}+\frac{y}{35}\leq 1$



Modulus func^n –

$$|x| = \begin{cases} -x, & x<0 \\ +x, & x\geq 0 \end{cases} =$$



eg $|5| = +(5) = +5$

$|-5| = -(-5) = +5$

Note –

① $|x|<a \Rightarrow -a<x<a$

② $|x|>a \Rightarrow x<-a$ or $x>a$

③ $|x-a|<l \Rightarrow a-l<x<a+l$

④ $|x-a|>l \Rightarrow x<a-l$ or $x>a+l$

Proof ① $|x|<a \Rightarrow \pm(x)<a$

Case I: $-x<a$

$x>-a$

Case II: $+x<a$

$x<a$



i.e $-a<x<a$

③ $|x-a|<l \Rightarrow \pm(x-a)<l$

Case I: $-(x-a)<l$

$x-a>-l$

$x>a-l$

Case II: $+(x-a)<l$

$x<a+l$

i.e. $a-l<x<a+l$

(148)

Eg: if $|4x-7|=5$ then value of $2|x|-|-x|=?$

Sol: $\pm(4x-7)=5$

$-(4x-7)=5$ & $+(4x-7)=5$

$-4x=-2$ & $4x=12$

$x=\frac{1}{2}$ & $x=3$

for $x=\frac{1}{2}$;

$2|x|-|-x| = 2\left|\frac{1}{2}\right|-\left|-\frac{1}{2}\right|$

$= 2\left(\frac{1}{2}\right)-\left(+\frac{1}{2}\right)$

$= 1-\frac{1}{2}=\frac{1}{2}$

for $x=3$;

$2|3|-|-3|$

$=2(+3)-(+3)$

$=6-3$

$=3$

Q: the Number of soln's of $x^2-5|x|+6=0$ is/are ? four soln's.

Soln: Case I: if $x>0$ then $|x|=x$

So $x^2-5|x|+6=0$

$x^2-5(+x)+6=0$

$x^2-5x+6=0$

$(x-2)(x-3)=0$

$x=2,3$

Case II: if $x<0$ then $|x|=-x$

$x^2-5(-x)+6=0$

$x^2+5x+6=0$

$(x+2)(x+3)=0$

$x=-2,-3$

Q. The Number of soln's of $x^2+|x|-6=0$ in case ? Two soln

Case I: if $x>0$ then $|x|=x$

$$x^2+(+x)-6=0$$
$$x^2+x-6=0$$
$$x^2+3x-2x-6=0$$
$$x(x+3)-2(x+3)=0$$
$$(x+3)(x-2)=0$$
$$x=-3,\ 2$$

-3 ✗ $\quad 2$ ✓

Case II: if $x<0$ then $|x|=-x$

$$x^2+(-x)-6=0$$
$$x^2-x-6=0$$
$$x^2-3x+2x-6=0$$
$$x(x-3)+2(x-3)=0$$
$$(x-3)(x+2)=0$$
$$\Rightarrow x=-2,\ 3$$

-2 ✓ $\quad 3$ ✗

Q. Solve; $1 \le |x-2| \le 3$

Sol: Case I:

$|x-2| \le 3 \Rightarrow 2-3 \le x \le 2+3$

i.e $-1 \le x \le 5$

Case II: $|x-2| \ge 1$

$\Rightarrow x \le 2-1$ or $x \ge 2+1$

$x \le 1$ or $x \ge 3$



So Ans is

$-1 \le x \le 1$ or $3 \le x \le 5$

$[-1,1] \cup [3,5]$

Q: Solve; $|x+1|+|x| > 3$ — (1)

Sol: we should focus on $x=0$ & $x=-1$

Case I: $x < -1 \Rightarrow (x+1) < 0$

By (1); $-(x+1)+(-x) > 3$

$-2x > 4 \Rightarrow \boxed{x < -2}$

Case II: $-1 < x < 0$ → $x+1 > 0$, $x < 0$

By (1), $+(x+1)+(-x) > 3$

$1 > 3$ Invalid

Case III: $x > 0 \Rightarrow x+1 > 0$

By (1); $+(x+1)+(+x) > 3$

$2x > 2 \Rightarrow \boxed{x > 1}$

ie $x < -2$ or $x > 1$



ie $(-\infty, -2) \cup (1, \infty)$


Thank
YOU